

Tiny_project

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2026-09-17

R Markdown

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```
Bank_Churn <- read.csv("C:\\Users\\gital\\OneDrive\\Desktop\\tiny project\\himanshu\\Bank_Churn.csv")
Bank_Churn
```

##	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance
## 1	15803976	Efremov	694	France	Female	31	10	0.00
## 2	15682541	Hartley	616	Spain	Female	36	6	132311.71
## 3	15695699	Calabrese	687	France	Male	35	8	0.00
## 4	15624188	Chiu	712	France	Female	33	6	0.00
## 5	15812191	Brennan	553	France	Male	33	4	118082.89
## 6	15636673	Onwuatiegwu	667	France	Male	31	1	119266.69
## 7	15594898	Hewitt	731	France	Male	43	2	0.00
## 8	15660211	Shih	629	Germany	Male	35	7	156847.29
## 9	15773972	Balashov	614	France	Male	50	4	137104.47
## 10	15746726	Doyle	438	Germany	Male	31	8	78398.69
## 11	15712287	Pokrovskii	652	France	Female	80	4	0.00
## 12	15702919	Collins	729	Germany	Male	30	6	63669.42
## 13	15674398	Russo	642	France	Male	38	3	0.00
## 14	15797960	Skinner	806	Germany	Female	59	0	135296.33
## 15	15631868	Robertson	744	Spain	Male	36	2	153804.44
## 16	15581539	Atkinson	474	Spain	Male	37	3	0.00
## 17	15662736	Doyle	559	France	Male	49	2	147069.78
## 18	15666252	Ritchie	706	Spain	Male	42	9	0.00
## 19	15677512	McEncroe	628	Spain	Female	22	3	0.00
## 20	15626114	Pearson	429	France	Male	24	4	95741.75
## 21	15810834	Gordon	525	Spain	Female	57	2	145965.33
## 22	15678910	Ts'ai	680	France	Female	30	8	141441.75
## 23	15694408	Lung	749	France	Male	40	1	139290.41
## 24	15585215	Yuan	763	France	Female	31	4	0.00
## 25	15682757	Pardey	734	France	Male	30	3	0.00
## 26	15736601	Tai	716	France	Male	35	4	144428.87
## 27	15601848	Scott	594	France	Male	35	2	0.00
## 28	15736008	Hunter	644	France	Female	46	9	95441.27
## 29	15669064	Mazzanti	671	Germany	Male	35	1	144848.74
## 30	15624528	L?	664	Germany	Male	26	7	116244.14
## 31	15598493	Beach	656	France	Male	50	7	0.00
## 32	15601274	Hsieh	667	Spain	Female	40	1	146502.07
## 33	15702669	Faulkner	663	Germany	Male	44	2	117028.60
## 34	15728669	Knowles	584	Germany	Female	30	8	112013.81
## 35	15742668	Day	626	Spain	Female	37	6	108269.37
## 36	15697441	Hsueh	485	France	Male	29	7	182123.79
## 37	15740476	Tsao	659	Germany	Female	32	3	150923.74
## 38	15648064	Kennedy	649	France	Male	33	2	0.00
## 39	15636624	Nwabugwu	805	Spain	Female	39	5	165272.13
## 40	15807923	Young	716	Germany	Female	39	10	115301.31
## 41	15745844	Kerr	642	Germany	Female	40	6	129502.49
## 42	15786170	Tien	659	France	Male	31	4	118342.26
## 43	15681081	Marrero	545	Spain	Female	47	5	0.00
## 44	15684484	White	543	France	Male	22	8	0.00
## 45	15785869	Pisano	718	France	Female	25	7	0.00
## 46	15763859	Brown	840	France	Female	43	7	0.00
## 47	15658935	Freeman	630	Germany	Female	34	9	106937.05
## 48	15747358	Russell	643	Germany	Male	59	3	170331.37
## 49	15735203	Seleznyov	654	Germany	Female	32	1	114510.85
## 50	15576256	Yusupova	582	France	Male	39	5	0.00
## 51	15659420	Foley	659	Spain	Male	32	3	107594.11
## 52	15593365	Shih	762	Spain	Male	39	2	81273.13
## 53	15777352	Ikedinachukwu	568	Spain	Female	32	7	169399.60
## 54	15812007	Power	670	Spain	Male	25	6	0.00

## 55	15625461	Amos	613	France	Female	45	1	187841.99
## 56	15739438	Reed	539	France	Male	30	0	0.00
## 57	15611759	Simmons	850	Spain	Female	57	8	126776.30
## 58	15661629	Ricci	522	Spain	Male	34	9	126436.29
## 59	15633950	Yen	737	France	Male	41	1	101960.74
## 60	15592386	Campbell	520	France	Male	58	3	0.00
## 61	15803716	West	706	Spain	Male	28	3	0.00
## 62	15696674	Robinson	643	Germany	Female	45	2	150842.93
## 63	15706365	Bianchi	648	France	Female	50	9	102535.57
## 64	15745088	Chen	443	Germany	Female	29	9	99027.61
## 65	15676715	Madukaego	640	France	Male	68	9	0.00
## 66	15613085	Ibrahimova	628	Spain	Female	33	3	0.00
## 67	15633537	Nolan	540	Germany	Female	42	9	87271.41
## 68	15594720	Scott	460	Germany	Female	35	8	102742.91
## 69	15684042	Blair	636	Germany	Male	34	2	40105.51
## 70	15583303	Monaldo	593	France	Female	29	2	152265.43
## 71	15611579	Sutherland	801	Spain	Male	42	4	141947.67
## 72	15774696	Cole	640	Germany	Female	75	1	106307.91
## 73	15694506	Briggs	611	Germany	Male	31	0	107884.81
## 74	15688074	Gregory	802	Germany	Male	31	1	125013.72
## 75	15759537	Bianchi	717	Germany	Male	35	7	58469.37
## 76	15758449	Angelo	769	France	Female	39	8	0.00
## 77	15583456	Gardiner	745	Germany	Male	45	10	117231.63
## 78	15667871	Kerr	572	Spain	Male	35	4	152390.26
## 79	15677371	Ko	629	Spain	Female	30	2	34013.63
## 80	15629677	Distefano	687	Spain	Female	39	2	0.00
## 81	15713578	Farrell	483	France	Female	50	9	0.00
## 82	15591509	Milano	690	France	Male	36	7	101583.11
## 83	15568240	Ting	492	Germany	Female	30	10	77168.87
## 84	15622993	Boyd	709	Germany	Male	28	8	124695.72
## 85	15689294	Onyemaechi	705	Germany	Male	44	3	105934.96
## 86	15720910	Black	560	France	Female	66	9	0.00
## 87	15721181	Oliver	611	Spain	Male	46	6	0.00
## 88	15776433	Greco	730	Spain	Male	62	2	0.00
## 89	15748936	Whitehead	709	Spain	Female	45	2	0.00
## 90	15717225	Ikemefuna	544	France	Female	21	10	161525.96
## 91	15685226	Morrison	712	Germany	Female	29	7	147199.07
## 92	15785611	Onyeoruru	752	Germany	Male	38	3	183102.29
## 93	15573456	Cunningham	648	Spain	Male	46	9	127209.00
## 94	15684548	Demidov	556	Spain	Male	38	8	0.00
## 95	15620505	Celis	594	Spain	Female	24	0	97378.54
## 96	15807432	Cheng	645	Germany	Female	37	2	136925.09
## 97	15584766	Knight	557	France	Male	33	3	54503.55
## 98	15612187	Morin	547	Germany	Male	32	8	155726.85
## 99	15762218	Mills	701	France	Female	39	9	0.00
## 100	15646372	Outhwaite	616	France	Female	66	1	135842.41
## 101	15690452	Tung	605	France	Male	52	1	63349.75
## 102	15747795	Pai	593	Germany	Female	38	4	129499.42
## 103	15781589	Carpenter	751	Spain	Male	52	8	0.00
## 104	15732674	Fennell	443	Spain	Male	36	6	70438.01
## 105	15642291	Fontaine	685	France	Male	23	8	0.00
## 106	15692761	Pratt	718	France	Male	36	9	0.00
## 107	15578045	Mitchell	538	Spain	Female	49	9	141434.04
## 108	15745354	Franklin	611	Spain	Female	37	4	0.00
## 109	15701376	K'ung	668	Germany	Male	37	10	152958.29
## 110	15691625	Ko	537	Germany	Female	41	3	138306.34

## 111	15566594	McKenzie	709	Spain	Male	23	10	0.00
## 112	15760431	Pino	850	France	Male	38	1	0.00
## 113	15686302	Fisk	745	Spain	Female	31	3	124328.84
## 114	15801559	Chiang	693	Germany	Female	41	9	181461.48
## 115	15810432	Moseley	795	Spain	Male	35	8	0.00
## 116	15809616	Hsiung	626	Spain	Male	26	8	0.00
## 117	15720559	Heath	487	Germany	Female	61	5	110368.03
## 118	15695632	Dellucci	556	France	Female	39	9	89588.35
## 119	15659843	Li	643	France	Female	46	6	0.00
## 120	15615624	De Salis	605	France	Female	28	6	0.00
## 121	15810418	T'ang	756	Germany	Female	60	3	115924.89
## 122	15716186	Richardson	586	France	Female	38	2	0.00
## 123	15674551	Fitch	535	Germany	Male	40	7	111756.50
## 124	15622834	Stevenson	678	France	Female	35	4	0.00
## 125	15566111	Estes	596	France	Male	39	9	0.00
## 126	15784597	Lattimore	648	France	Male	26	9	162923.85
## 127	15652883	Chung	492	Germany	Male	39	10	124576.65
## 128	15806964	Utz	702	France	Male	45	0	80793.58
## 129	15576313	Wei	486	Germany	Female	40	9	71340.09
## 130	15806467	Boyle	568	Germany	Male	40	1	99282.63
## 131	15597602	Nwachinemelu	619	Germany	Male	57	3	137946.39
## 132	15743040	Kuznetsova	724	Germany	Male	41	2	127892.57
## 133	15705521	Pisani	548	Germany	Female	33	0	101084.36
## 134	15595039	Manna	545	Germany	Female	37	8	114754.08
## 135	15799384	Collier	683	France	Male	33	8	0.00
## 136	15581197	Ricci	762	France	Female	51	3	99286.98
## 137	15693737	Carr	627	Germany	Female	30	4	79871.02
## 138	15624623	Hs?	516	France	Male	35	10	104088.59
## 139	15783501	Findlay	800	France	Female	38	2	168190.33
## 140	15690134	Hughes	464	Germany	Female	42	3	85679.25
## 141	15782735	Chukwuemeka	626	France	Female	35	3	0.00
## 142	15611088	Genovese	790	France	Female	31	9	0.00
## 143	15672145	Swift	534	France	Female	34	7	121551.58
## 144	15732628	Ugoji	745	France	Male	46	2	122220.19
## 145	15787470	Parkinson	553	Spain	Male	47	3	116528.15
## 146	15803406	Ross	748	France	Female	26	1	77780.29
## 147	15730460	Oleary	722	France	Male	37	2	0.00
## 148	15644572	Turnbull	501	France	Male	40	4	125832.20
## 149	15694860	Uspensky	675	France	Female	38	6	68065.80
## 150	15658169	Cook	778	Spain	Female	47	6	127299.34
## 151	15794396	Newbold	494	Germany	Female	38	7	174937.64
## 152	15785798	Uchechukwu	850	France	Male	40	9	0.00
## 153	15710825	Ch'en	592	Spain	Male	31	7	110071.10
## 154	15668444	He	590	Spain	Female	44	3	139432.37
## 155	15726631	Hilton	758	France	Female	39	6	127357.76
## 156	15733797	Sal	506	France	Male	36	5	0.00
## 157	15747960	Eluemuno	733	France	Male	33	3	0.00
## 158	15634632	Titus	711	France	Male	38	3	0.00
## 159	15707362	Yin	514	Germany	Male	43	1	95556.31
## 160	15662976	Lettiere	637	Spain	Male	37	8	0.00
## 161	15732778	Templeman	468	Germany	Male	29	1	111681.98
## 162	15718443	Chibuzo	539	France	Male	39	3	0.00
## 163	15670039	Sun	509	Spain	Female	25	3	108738.71
## 164	15773792	Evans	662	France	Female	32	4	133950.37
## 165	15613786	Ogbonnaya	818	Spain	Male	26	4	0.00
## 166	15726032	Enyinnaya	608	France	Male	33	9	89968.69

## 167	15663252	Olisanugo	850	Spain	Female	32	9	0.00
## 168	15593782	Brookes	816	Germany	Female	38	5	130878.75
## 169	15633283	Padovano	536	France	Male	35	8	0.00
## 170	15749167	Fisk	753	France	Male	35	3	0.00
## 171	15759298	Shih	631	Spain	Male	27	10	134169.62
## 172	15683625	Hare	703	France	Male	37	1	149762.08
## 173	15635367	Muir	774	France	Male	26	2	93844.69
## 174	15681705	Fanucci	785	France	Male	28	8	0.00
## 175	15603156	Elewechi	571	France	Female	33	1	0.00
## 176	15591986	Johnston	621	Germany	Male	46	6	141078.37
## 177	15798888	Pisano	605	Germany	Female	31	1	117992.59
## 178	15809722	Ankudinov	611	France	Female	40	8	100812.33
## 179	15677538	Nwokike	569	France	Male	38	7	0.00
## 180	15797736	Smith	658	France	Male	29	4	80262.60
## 181	15695585	Atkins	788	Spain	Male	34	6	156478.62
## 182	15744398	Burns	525	France	Female	23	5	0.00
## 183	15750658	Obiuto	798	France	Male	37	8	0.00
## 184	15578186	Pirozzi	486	Germany	Male	37	9	115217.99
## 185	15676519	George	615	Spain	Male	61	9	0.00
## 186	15637954	Lewis	730	France	Female	35	0	155470.55
## 187	15758639	Moran	641	France	Male	37	7	0.00
## 188	15613772	Dalrymple	542	France	Male	39	3	135096.77
## 189	15731744	Carslaw	692	France	Male	30	2	0.00
## 190	15807709	Kirby	714	Germany	Female	55	9	180075.22
## 191	15714689	Houghton	591	Spain	Male	29	1	97541.24
## 192	15699005	Martin	710	France	Female	41	2	156067.05
## 193	15624170	Tan	639	France	Female	38	4	81550.94
## 194	15725679	Hsia	531	France	Female	47	6	0.00
## 195	15585865	Westerberg	673	France	Female	38	2	170061.92
## 196	15804256	Hale	765	Germany	Male	36	8	92310.54
## 197	15662403	Kryukova	622	France	Female	32	6	169089.38
## 198	15733616	Sopuluchukwu	806	France	Male	40	5	80613.93
## 199	15591995	Barry	757	Germany	Male	26	8	121581.56
## 200	15677020	Selezneva	570	France	Female	58	8	0.00

##	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
## 1	2	1	0	160990.27	0
## 2	1	0	0	15462.84	0
## 3	2	1	0	10334.05	0
## 4	2	1	1	190686.16	0
## 5	1	0	0	94440.45	0
## 6	1	1	1	28257.63	0
## 7	1	1	1	170034.95	1
## 8	2	1	0	31824.29	0
## 9	1	1	0	127166.49	1
## 10	1	1	0	44937.01	0
## 11	2	1	1	188603.07	0
## 12	1	1	0	145111.37	0
## 13	2	0	0	171463.83	0
## 14	1	1	0	182822.50	0
## 15	1	1	1	87213.33	0
## 16	2	0	0	57175.32	0
## 17	1	1	0	120540.83	1
## 18	2	1	1	28714.34	0
## 19	1	1	0	85426.28	0
## 20	1	1	0	46170.75	0
## 21	1	1	1	64448.36	0

## 22	1	1	1	16278.97	0
## 23	1	1	0	182855.42	1
## 24	2	0	0	50404.72	0
## 25	2	1	0	107640.25	0
## 26	1	1	0	134132.65	0
## 27	2	1	0	103480.69	0
## 28	1	1	0	108761.05	1
## 29	1	1	1	179012.30	0
## 30	2	1	1	95145.14	0
## 31	2	0	1	72143.44	0
## 32	1	1	0	19162.89	0
## 33	2	0	1	144680.18	0
## 34	1	1	0	177772.03	1
## 35	1	1	0	5597.94	0
## 36	1	1	0	116828.51	1
## 37	2	0	1	174652.51	0
## 38	2	1	0	2010.98	0
## 39	1	1	0	14109.85	1
## 40	1	1	0	43527.40	1
## 41	2	0	1	86099.23	1
## 42	1	0	0	161574.19	0
## 43	2	1	1	38970.14	0
## 44	2	0	0	127587.22	0
## 45	2	1	0	30380.12	0
## 46	2	1	0	90908.95	0
## 47	2	1	0	138275.01	0
## 48	1	1	1	32171.79	0
## 49	1	1	1	126143.23	0
## 50	2	1	1	129892.93	0
## 51	2	1	1	102416.84	0
## 52	1	1	1	18719.67	0
## 53	1	1	0	61936.22	0
## 54	2	1	1	78358.94	0
## 55	2	1	1	147224.27	0
## 56	2	1	0	160979.66	0
## 57	2	1	1	132298.49	0
## 58	1	1	0	174248.52	1
## 59	1	1	1	123547.28	0
## 60	2	0	1	32790.02	0
## 61	2	0	1	181543.67	0
## 62	1	0	1	2319.96	1
## 63	1	1	1	189543.19	0
## 64	2	1	0	10940.40	0
## 65	2	1	1	199493.38	0
## 66	1	1	1	188193.25	0
## 67	2	1	0	172572.64	0
## 68	2	1	1	189339.60	0
## 69	2	0	1	53512.16	0
## 70	1	1	0	34004.44	0
## 71	1	1	1	10598.29	0
## 72	2	0	1	113428.77	0
## 73	2	1	1	183487.98	0
## 74	1	1	1	187658.09	0
## 75	2	1	1	172459.39	0
## 76	1	0	1	21016.00	0
## 77	3	1	1	122381.02	1

## 78	1	1	0	128123.66	0
## 79	1	1	0	19570.63	0
## 80	3	0	0	188150.60	1
## 81	2	1	1	111020.24	0
## 82	2	1	0	123775.15	0
## 83	2	0	1	146700.22	0
## 84	2	1	0	145251.35	0
## 85	1	1	0	82463.69	0
## 86	1	1	1	15928.49	0
## 87	2	1	0	45886.33	0
## 88	2	1	1	186489.95	0
## 89	2	0	1	162922.65	0
## 90	2	1	0	9262.77	0
## 91	1	1	1	84932.40	0
## 92	1	1	1	71557.12	0
## 93	2	1	0	77405.95	1
## 94	2	0	0	417.41	1
## 95	1	1	1	71405.17	0
## 96	2	0	1	153400.24	0
## 97	1	1	1	371.05	0
## 98	1	1	0	67789.99	0
## 99	2	0	1	145894.90	0
## 100	1	1	0	183840.51	1
## 101	1	1	0	108887.44	0
## 102	1	1	1	154071.27	0
## 103	2	0	1	179291.85	0
## 104	2	0	1	56937.43	0
## 105	2	1	1	112239.03	0
## 106	1	1	0	45909.87	0
## 107	1	0	0	173779.25	1
## 108	2	1	0	125696.26	0
## 109	2	1	1	159585.61	0
## 110	1	1	0	106761.47	0
## 111	2	0	0	129590.18	0
## 112	2	1	1	80006.65	0
## 113	1	1	1	140451.52	0
## 114	3	1	1	187929.43	1
## 115	2	1	0	167155.36	0
## 116	2	0	0	191420.71	0
## 117	1	0	0	11384.45	1
## 118	1	1	1	94898.10	0
## 119	2	0	0	106781.59	0
## 120	2	0	0	159508.52	0
## 121	1	1	0	93524.19	1
## 122	2	1	0	87168.46	0
## 123	1	1	0	8128.32	1
## 124	1	1	0	125518.32	0
## 125	1	1	0	48963.59	0
## 126	1	1	0	98368.24	0
## 127	2	1	0	148584.61	0
## 128	1	1	1	27474.81	0
## 129	1	1	0	76192.21	0
## 130	1	0	0	134600.94	1
## 131	1	1	1	72467.99	1
## 132	2	0	1	199645.45	0
## 133	1	1	0	42749.85	0

## 134	1	1	0	136050.44	1
## 135	1	0	0	73564.44	0
## 136	1	0	1	85578.63	0
## 137	2	1	0	129826.89	0
## 138	2	0	0	119666.00	0
## 139	2	1	0	68052.08	0
## 140	1	1	1	164104.74	0
## 141	1	0	0	80190.36	0
## 142	2	1	0	84126.75	0
## 143	2	1	1	70179.00	0
## 144	1	1	1	118024.10	0
## 145	1	0	0	145704.19	1
## 146	1	0	1	183049.41	0
## 147	1	0	0	120906.83	0
## 148	1	1	1	100433.83	0
## 149	1	0	0	138777.00	1
## 150	2	1	0	124694.99	0
## 151	1	1	0	40084.32	0
## 152	2	0	1	119232.33	0
## 153	1	0	0	43921.36	0
## 154	1	1	0	62222.81	0
## 155	1	0	1	56577.00	0
## 156	2	1	0	164253.35	0
## 157	1	1	1	7666.73	0
## 158	2	1	0	68487.51	0
## 159	1	0	1	199273.98	1
## 160	1	1	1	186062.36	0
## 161	2	1	1	195711.16	0
## 162	2	1	0	36692.17	0
## 163	2	1	0	106920.57	0
## 164	1	1	1	48725.68	1
## 165	2	1	1	167036.94	0
## 166	1	1	0	68777.26	0
## 167	2	1	1	18924.92	0
## 168	3	1	0	71905.77	1
## 169	2	1	0	64833.28	0
## 170	2	1	1	184843.77	0
## 171	1	1	1	176730.02	0
## 172	1	1	0	20629.40	1
## 173	1	1	0	28415.36	0
## 174	2	1	0	77231.27	0
## 175	2	1	0	102750.70	0
## 176	1	0	0	34580.80	1
## 177	1	1	1	183598.77	0
## 178	2	1	0	147358.27	0
## 179	1	1	1	108469.20	0
## 180	1	1	1	20612.82	0
## 181	1	0	1	181196.76	0
## 182	2	1	0	160249.10	0
## 183	3	0	0	110783.28	0
## 184	2	1	0	144995.33	0
## 185	2	1	0	150227.85	1
## 186	1	1	1	53718.28	0
## 187	2	1	0	75248.30	0
## 188	1	1	1	14353.43	1
## 189	2	0	1	130486.57	0

## 190	1	1	1	100127.71	0
## 191	1	1	1	196356.17	0
## 192	1	1	1	9983.88	0
## 193	2	0	1	118974.77	0
## 194	1	0	0	194998.34	1
## 195	2	0	0	134901.34	1
## 196	2	1	1	72924.56	0
## 197	2	1	0	101057.95	0
## 198	1	1	1	142838.64	0
## 199	2	1	1	127059.04	0
## 200	1	0	1	116503.92	1

```
head(Bank_Churn)
```

##	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance
## 1	15803976	Efremov	694	France	Female	31	10	0.0
## 2	15682541	Hartley	616	Spain	Female	36	6	132311.7
## 3	15695699	Calabrese	687	France	Male	35	8	0.0
## 4	15624188	Chiu	712	France	Female	33	6	0.0
## 5	15812191	Brennan	553	France	Male	33	4	118082.9
## 6	15636673	Onwuatiegwu	667	France	Male	31	1	119266.7

##	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
## 1	2	1	0	160990.27	0
## 2	1	0	0	15462.84	0
## 3	2	1	0	10334.05	0
## 4	2	1	1	190686.16	0
## 5	1	0	0	94440.45	0
## 6	1	1	1	28257.63	0

```
summary(Bank_Churn)
```

```
##      CustomerId      Surname      CreditScore      Geography
## Min.   :15566111 Length   :200 Min.    :429.0 Length  :200
## 1st Qu.:15633847 N.unique :190 1st Qu.:568.8 N.unique : 3
## Median :15691039 N.blank  : 0 Median :643.0 N.blank  : 0
## Mean   :15692776 Min.nchar: 2 Mean   :644.3 Min.nchar: 5
## 3rd Qu.:15746065 Max.nchar:13 3rd Qu.:716.0 Max.nchar: 7
## Max.   :15812191 Max.    :850.0
##      Gender      Age      Tenure      Balance
## Length   :200 Min.    :21.00 Min.    : 0.00 Min.    : 0
## N.unique  : 2 1st Qu.:31.75 1st Qu.: 2.00 1st Qu.: 0
## N.blank   : 0 Median :37.00 Median : 5.00 Median : 98284
## Min.nchar: 4 Mean   :38.22 Mean   : 5.07 Mean   : 77942
## Max.nchar: 6 3rd Qu.:42.00 3rd Qu.: 8.00 3rd Qu.:129500
## Max.      :80.00 Max.    :10.00 Max.    :187842
## NumOfProducts HasCrCard IsActiveMember EstimatedSalary
## Min.    :1.000 Min.    :0.000 Min.    :0.00 Min.    : 371.1
## 1st Qu.:1.000 1st Qu.:0.000 1st Qu.:0.00 1st Qu.: 55862.3
## Median :1.000 Median :1.000 Median :0.00 Median :108054.7
## Mean   :1.515 Mean   :0.735 Mean   :0.47 Mean   :103988.6
## 3rd Qu.:2.000 3rd Qu.:1.000 3rd Qu.:1.00 3rd Qu.:151020.9
## Max.    :3.000 Max.    :1.000 Max.    :1.00 Max.    :199645.5
##      Exited
## Min.    :0.000
## 1st Qu.:0.000
## Median :0.000
## Mean    :0.185
## 3rd Qu.:0.000
## Max.    :1.000
```

```
str(Bank_Churn)
```

```
## 'data.frame': 200 obs. of 13 variables:
## $ CustomerId : int 15803976 15682541 15695699 15624188 15812191 15636673 15594898 1566
0211 15773972 15746726 ...
## $ Surname : chr "Efremov" "Hartley" "Calabrese" "Chiu" ...
## $ CreditScore : int 694 616 687 712 553 667 731 629 614 438 ...
## $ Geography : chr "France" "Spain" "France" "France" ...
## $ Gender : chr "Female" "Female" "Male" "Female" ...
## $ Age : int 31 36 35 33 33 31 43 35 50 31 ...
## $ Tenure : int 10 6 8 6 4 1 2 7 4 8 ...
## $ Balance : num 0 132312 0 0 118083 ...
## $ NumOfProducts : int 2 1 2 2 1 1 1 2 1 1 ...
## $ HasCrCard : int 1 0 1 1 0 1 1 1 1 1 ...
## $ IsActiveMember : int 0 0 0 1 0 1 1 0 0 0 ...
## $ EstimatedSalary: num 160990 15463 10334 190686 94440 ...
## $ Exited : int 0 0 0 0 0 0 1 0 1 0 ...
```

```
colSums(is.na(Bank_Churn))
```

##	CustomerId	Surname	CreditScore	Geography	Gender
##	0	0	0	0	0
##	Age	Tenure	Balance	NumOfProducts	HasCrCard
##	0	0	0	0	0
##	IsActiveMember	EstimatedSalary	Exited		
##	0	0	0		

```
median(Bank_Churn$Balance)
```

```
## [1] 98284.43
```

```
var(Bank_Churn$Balance)
```

```
## [1] 4003826295
```

```
sd(Bank_Churn$Balance)
```

```
## [1] 63275.8
```

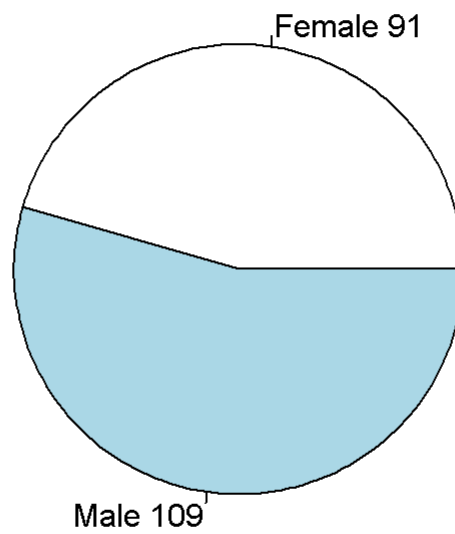
```
gender_count <- table(Bank_Churn$Gender)
```

```
gender_count
```

```
##  
## Female    Male  
##      91     109
```

```
pie(  
  gender_count,  
  main = "Gender Distribution",  
  labels = paste(  
    names(gender_count),  
    gender_count  
  )  
)
```

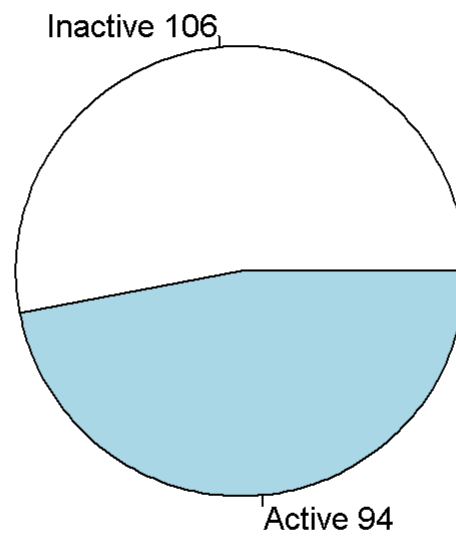
Gender Distribution



```
result_count <- table(Bank_Churn$IsActiveMember)
names(result_count) <- c("Inactive", "Active")
```

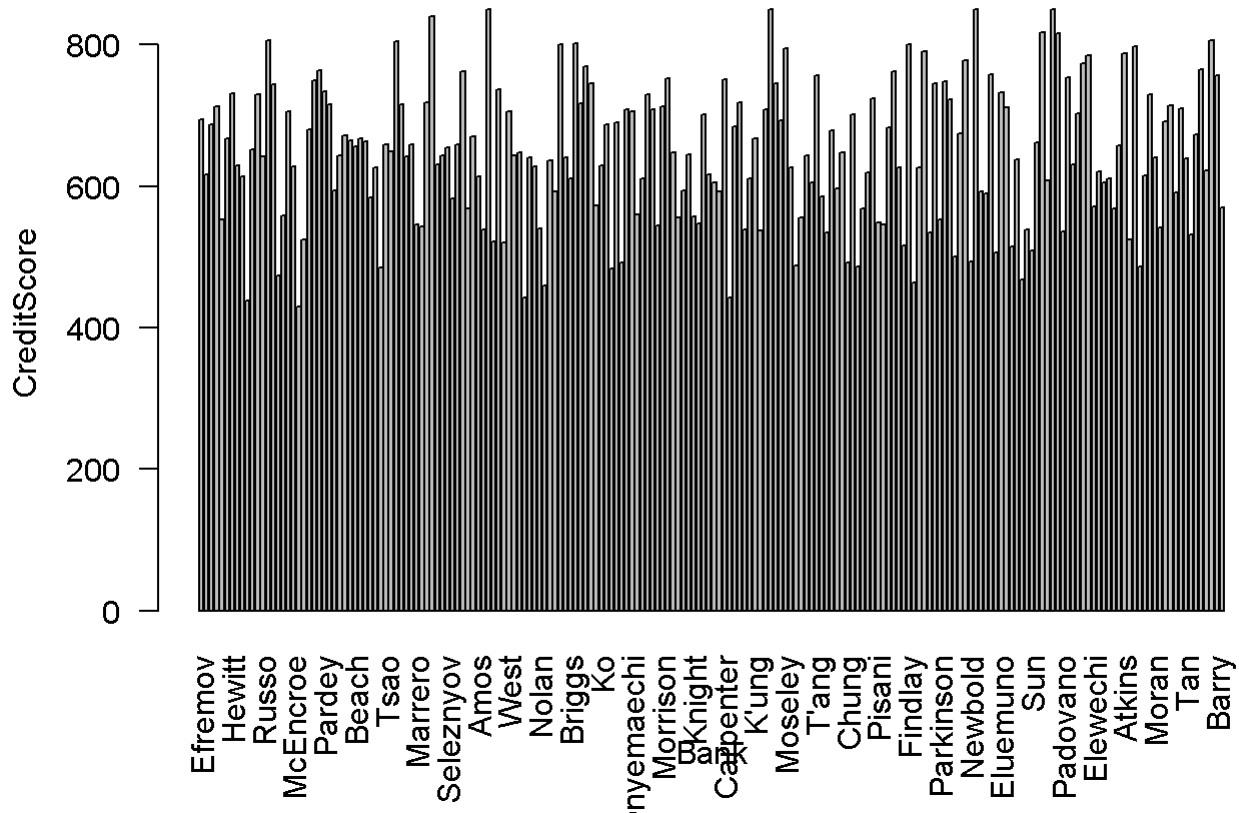
```
pie(
  result_count,
  main = "Bank Member Distribution",
  labels = paste(
    names(result_count),
    result_count
  )
)
```

Bank Member Distribution



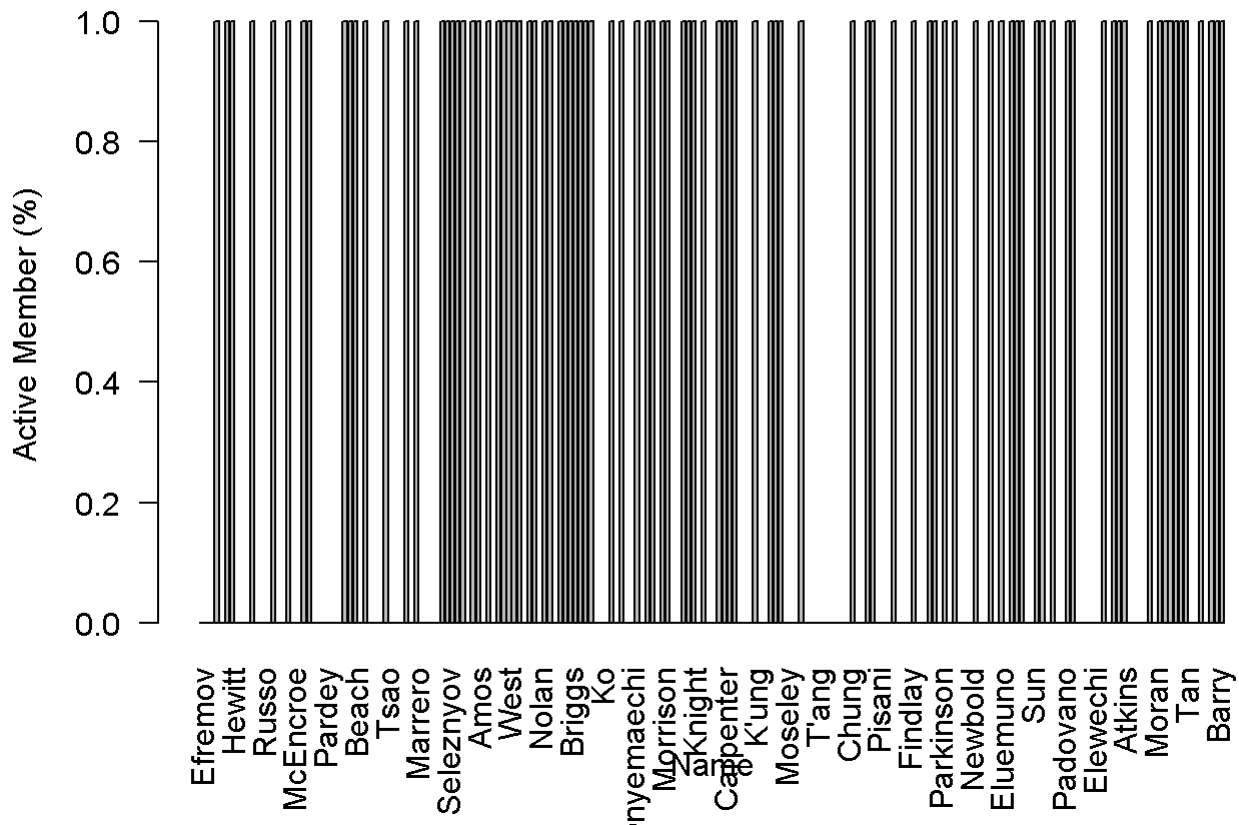
```
barplot(  
  Bank_Churn$CreditScore,  
  names.arg = Bank_Churn$Surname,  
  main = "Bank CreditScore",  
  xlab = "Bank",  
  ylab = "CreditScore",  
  las = 2  
)
```

Bank CreditScore



```
barplot(
  Bank_Churn$I,
  names.arg = Bank_Churn$Surname,
  main = "Bank Active member",
  xlab = "Name",
  ylab = "Active Member (%)",
  las = 2
)
```

Bank Active member



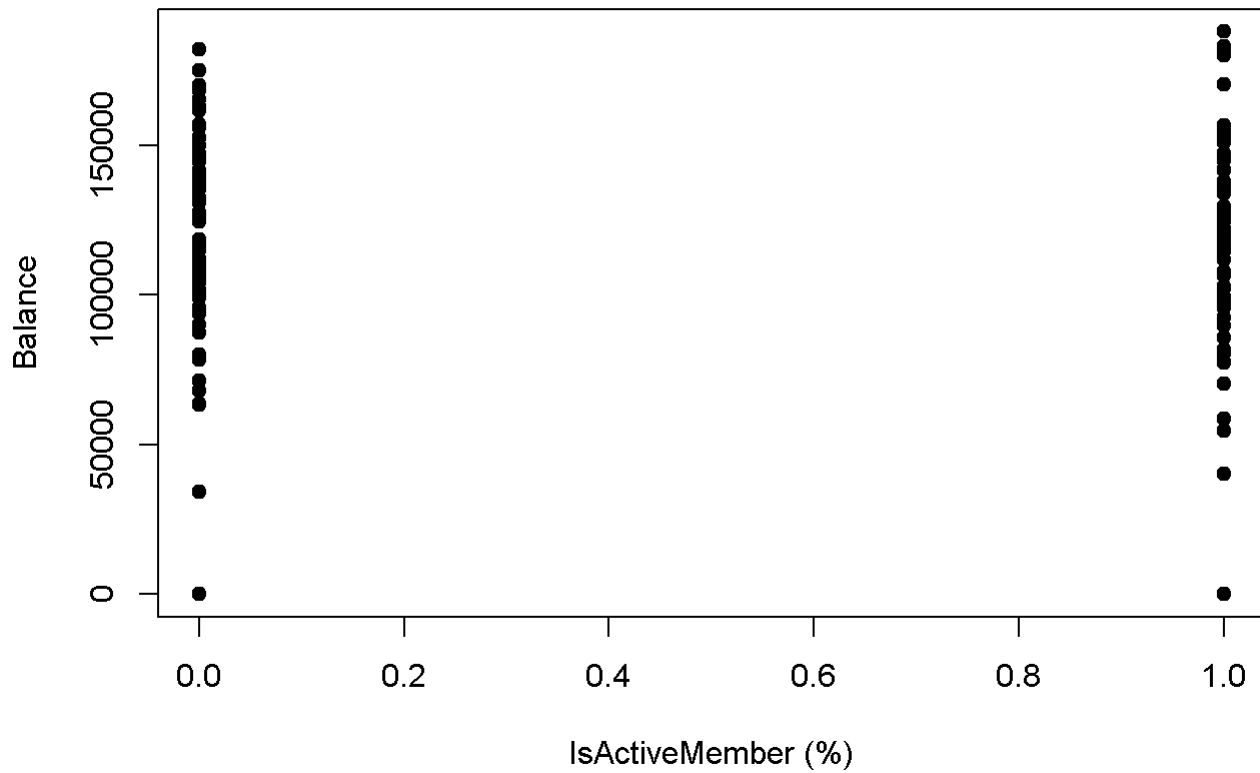
```
cor(
  Bank_Churn$IsActiveMember,
  Bank_Churn$Balance
)
```

```
## [1] 0.05032984
```

Including Plots

You can also embed plots, for example:

IsActiveMember vs Balance



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.